

Coding-Layout, Readability and Reusability

Date	8 July 2026
Team ID	SWETID-2026-5556
Project name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This section evaluates the structure, documentation, and modularity of the source code. It ensures that standard programming practices are followed to keep the system maintainable and scalable.

Code-Layout Checklist:

S No	Code Quality Parameter	Description	Followed (Yes/No/partial)	Remark
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Standard PEP 8 configuration (4 spaces) applied across all .py files and notebooks.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders created for data/, models/, src/app/, and notebooks.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Clear data science naming used (e.g., credit-score, annual-income, is-approved).
4	Function / Method Names	Names Functions are descriptively named	Yes	Snake-case used for actions (e.g., preprocess-data (), predict-default-risk ()).
5	Code Comments	Inline and block comments explain logic	Yes	Complex ML steps, hyperparameter tuning, and data leaks prevention are well-documented.
6	Modular Design	Code is split into reusable functions/modules	Yes	Data scaling and training logic split into dedicated utility scripts.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Cleaned up scratchpad notebook cells and kept only production pipeline code.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Try-except blocks handle missing dataset fields and API communication faults.

Reusable Components/Modules:

S No	Component /module Name	Language/ Technology	Where Reused	Reusability Level (High/Medium/ /Low)
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1	data_preprocessing.py	Python/ Pandas	Reused in both the model training script and the real-time inference Flask application pipeline	High
2	model_loader.py	Python / Joblib	Reused by core logic microservices to load the pre-trained evaluation model weights into memory.	High
3	sms_notifier.py	Python / Twilio API	Reused across multiple status update endpoints to push automated message results to clients.	Medium

Coding-Layout, Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Repository follows the ideal clean architecture template separating frontend view layers from backend calculation engines.
Readability	4	Clean variable declarations paired with robust inline comments make navigating the data science notebook highly intuitive.
Reusability	5	Main functional layers (data transforms and predictive model triggers) operate entirely out of independent utilities.
Documentation / Comments	4	Every analytical code block includes descriptive annotations highlighting expected tensor shapes and data frame behaviours.
Overall Score	4.5	The system codebase is exceptionally well-structured, easy to scale, and thoroughly checked against manual runtime faults.